

Hands-On Patient Profile Visualization

Refining the R package `patientProfilesVis`
with `ggplot2`, Interacting in shiny

KuenHung Lin @ 01Oct2026



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About me



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About you

Please share in the chat ...

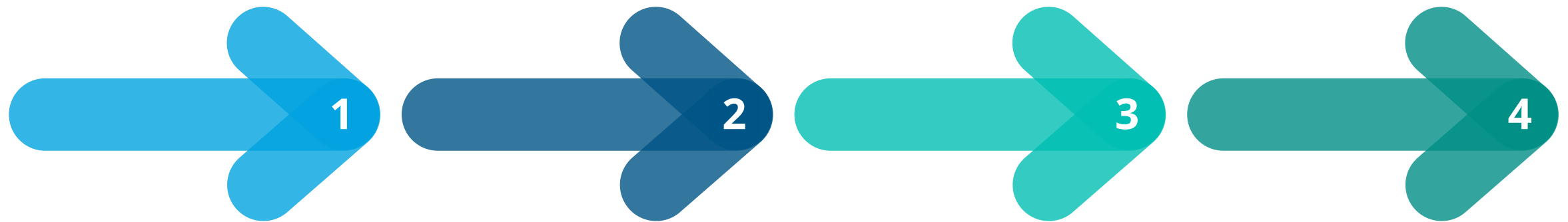
- Your name
- Your professional affiliation
- Have you created patient profile visualizations in R before
- Have you used the patientProfilesVis package before

Learning Philosophy

Learn new tools by solving real problems

Goal

Practice



Packages

Outcome

Today's Workshop Journey

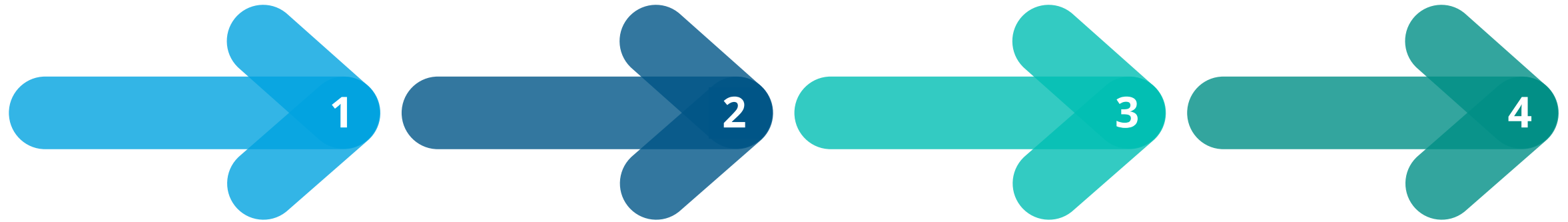
Applying the learning philosophy in practice

Goal

Patient Profile Visualization
What it is and why it matters

Practice

Build, Customize, and Extend
Learn through hands-on exercises



Packages

patientProfilesVis · ggplot2 · shiny
Leverage existing solutions

Outcome

Learn Faster, Solve More
Adapt existing tools to real-world challenges

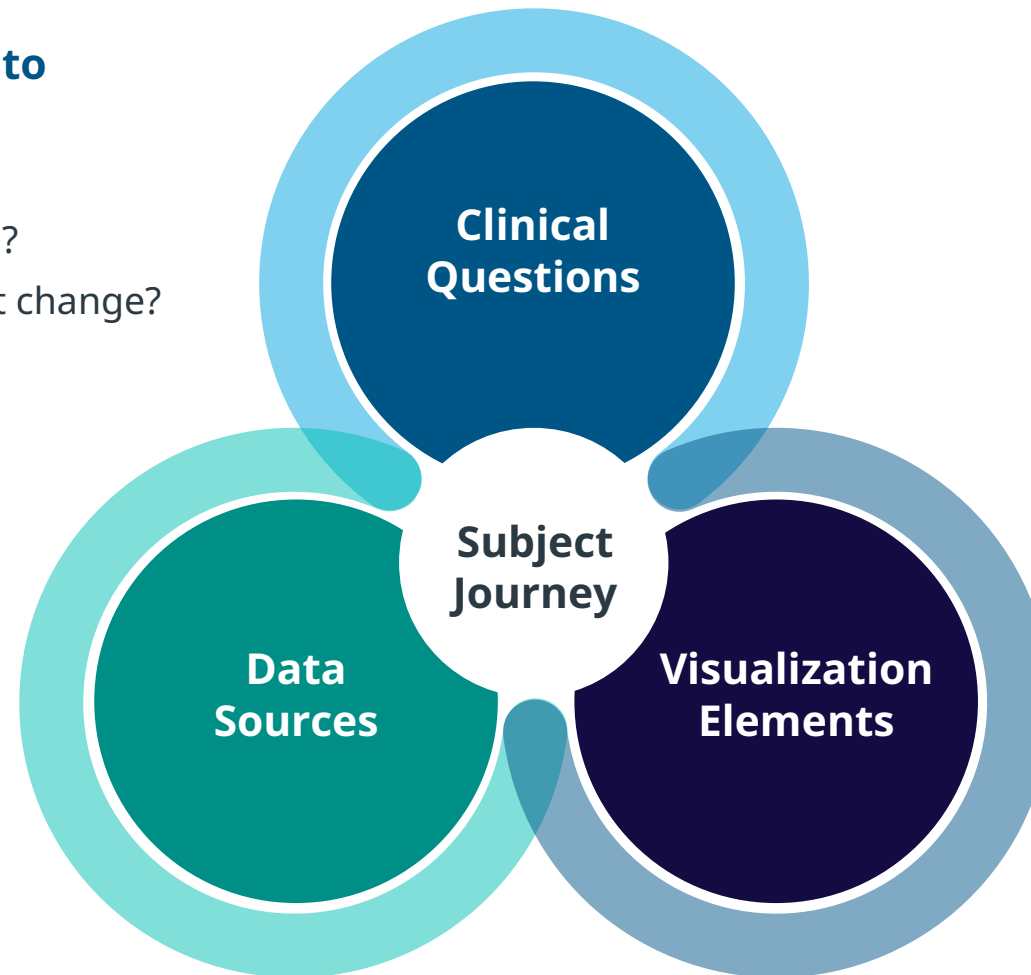
Understanding Patient Profiles

What do we want to understand?

- What happened?
- When did it happen?
- How did the subject change?

What data can we use?

- Raw Data
- SDTM
- ADaM

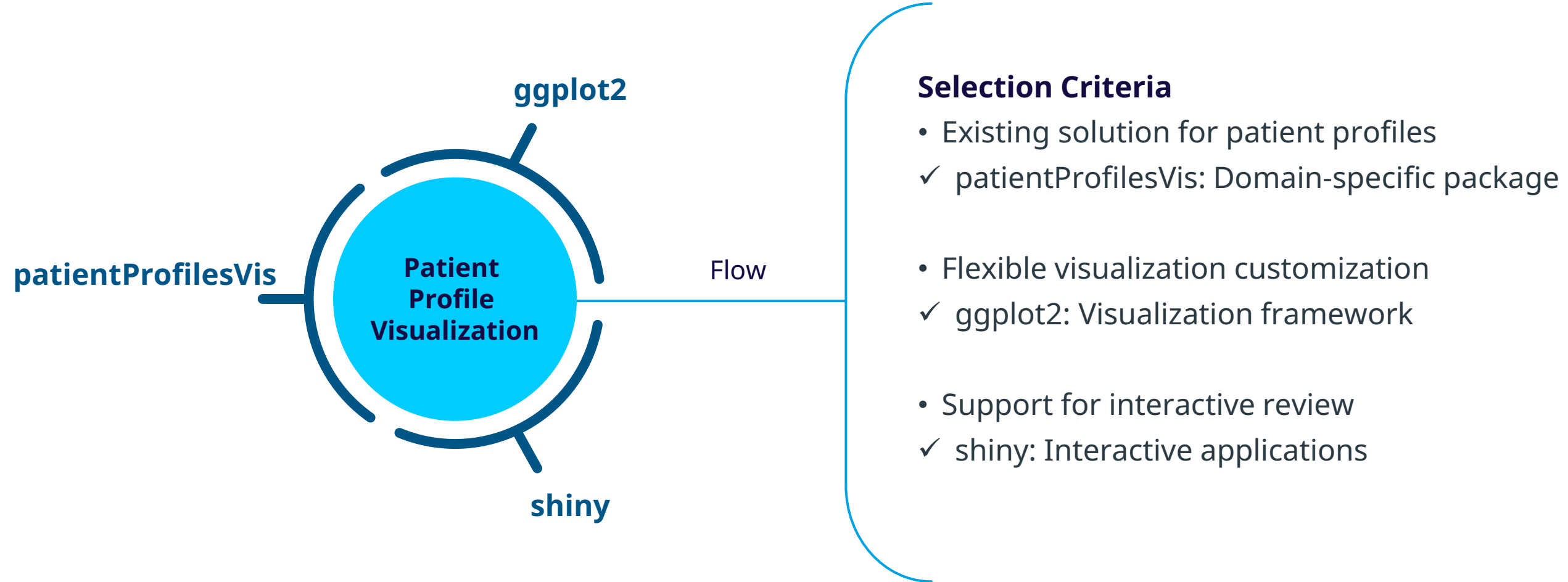


How can we present it?

- Events
- Intervals
- Trends

From Goal to Packages

Selecting existing solutions for patient profile visualization



How the Packages Work Together

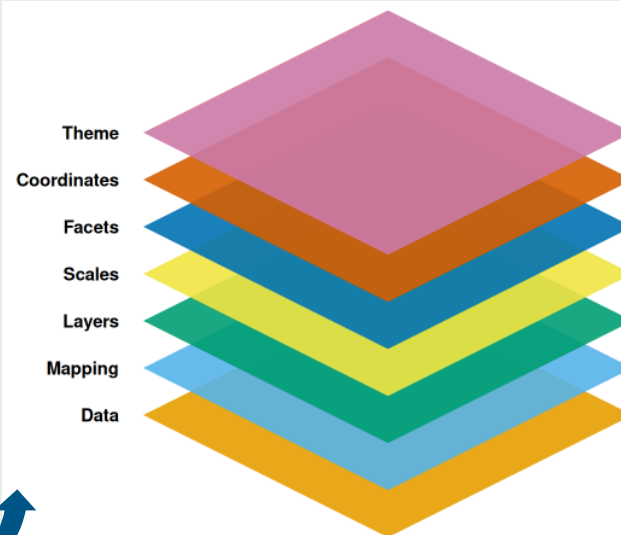
Generate with patientProfilesVis, customize with ggplot2, and extend with Shiny

Different types of visualization (a.k.a 'modules') are available via dedicated R function. Each function creates a separate visualization for each subject available in the dataset.

Four plot types/modules are currently available in the package:

- **'text'** module: patient specific information formatted as text, available via the `subjectProfileTextPlot` function
- **'interval'** module: representation of event with a start and end time, available via the `subjectProfileIntervalPlot` function
- **'event'** module: representation of event occurring at a single time, available via the `subjectProfileEventPlot` function
- **'line'** module: representation of the evolution of a continuous parameter across time via the `subjectProfileLinePlot` function

Each of this function returns a nested list of plots (`ggplot` object).



```
ui <- fluidPage(  
  plotOutput("plot", click = "plot_click"),  
  tableOutput("data")  
)  
server <- function(input, output, session) {  
  output$plot <- renderPlot({  
    ggplot(mtcars, aes(wt, mpg)) + geom_point()  
  }, res = 96)  
  
  output$data <- renderTable({  
    req(input$plot_click)  
    nearPoints(mtcars, input$plot_click)  
  })  
}
```

Generate Patient Profile Modules

Create reusable profile components through task-specific functions.

Text, interval, event, and line modules provide a structured starting point for patient profile visualization.

Source: [patientProfilesVis Introduction Vignette \(CRAN\)](#)

Customize Through Graphical Layers

Patient profile modules are returned as ggplot2 objects.

Layers, scales, facets, and themes can be adjusted to meet study-specific visualization needs.

Source: [Introduction to ggplot2 \(ggplot2 Documentation\)](#)

Extend to Interactive Review

Integrate customized visualizations into Shiny applications.

Enable subject selection, dynamic updates, and interactive data exploration.

Source: [Mastering Shiny by Hadley Wickham](#)

Setup for Hands-On Exercises

- Lock file built with Windows R 4.5.1
- Packages used in this workshop
 - patientProfilesVis (v2.0.10)
Installs the following dependencies automatically:
 - › clinUtils
 - › ggplot2
 - › cowplot
 - dplyr (any recent version)
 - shiny (any recent version)

Get the Workshop Materials

1. Open the GitHub repository
[liinhater/rpharma-2026-ppvis-workshop](https://github.com/liinhater/rpharma-2026-ppvis-workshop)
2. Download and extract
Select Code → Download ZIP, then extract it locally
3. Set up the environment
Refer to the README for setup instructions

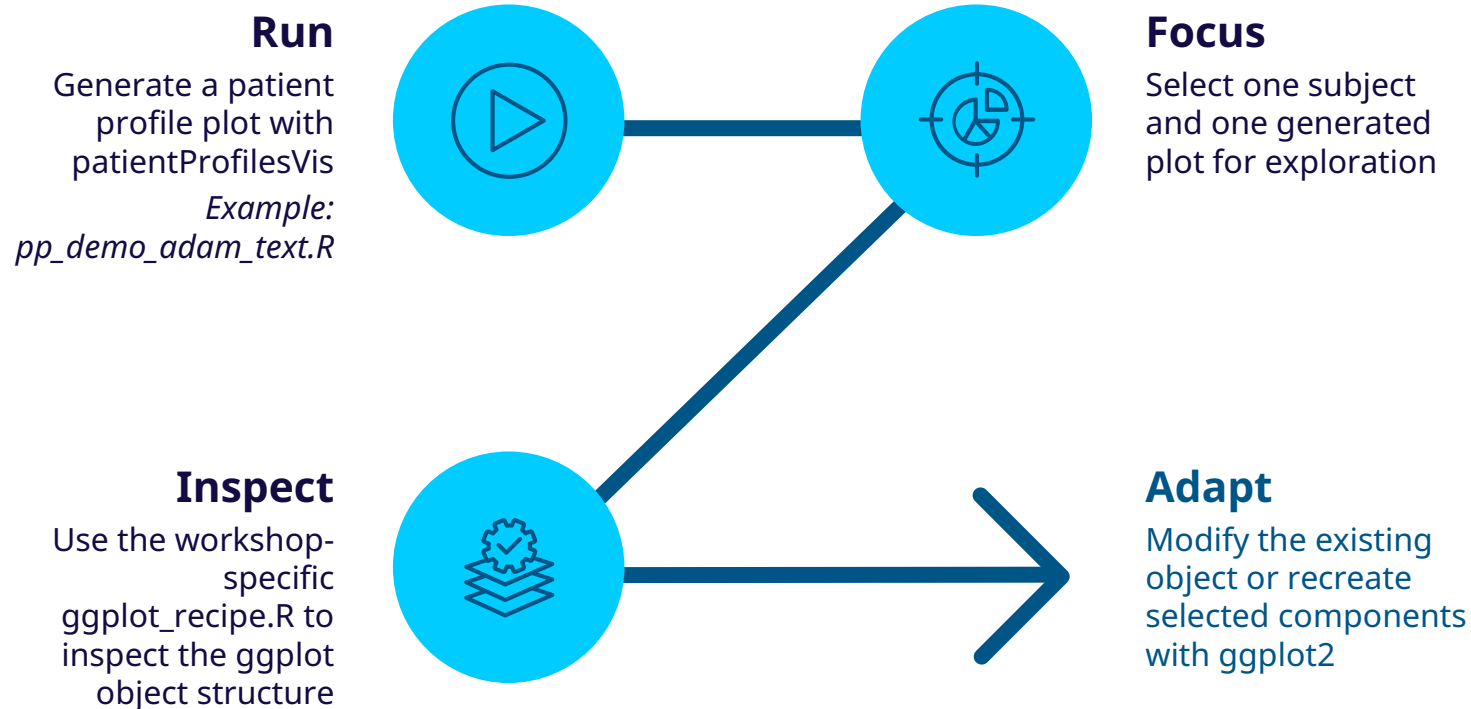
Estimated Setup Time



~5 min

How We Will Explore the Examples

Run, focus, inspect, and adapt



Workshop Focus

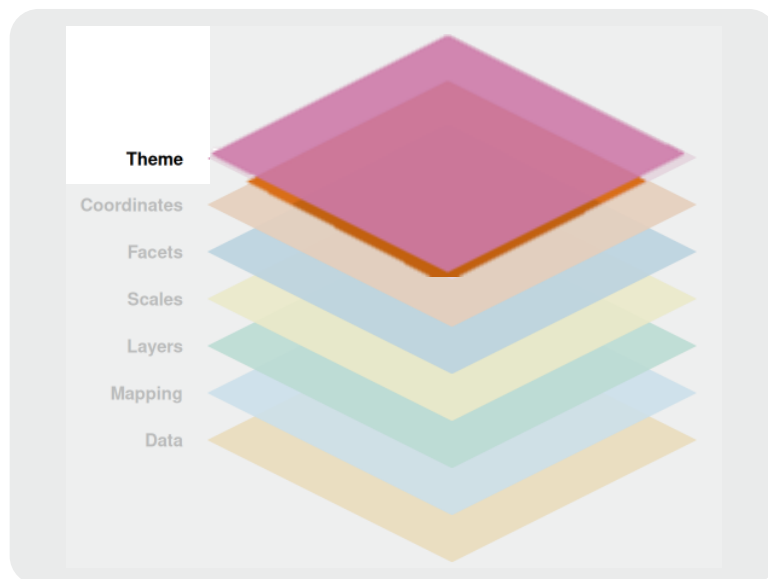
- ADaM-based examples
- One `patientProfilesVis` function at a time
- One subject and one generated plot
- Understanding and customizing `ggplot` objects

Out of Scope

- PDF report generation
- Pagination and multi-page outputs

THEME

Controls the overall appearance of a ggplot



Theme Layer

Theme controls non-data elements such as backgrounds, grid lines, fonts, axes, and legends.

It serves as a foundation for plot styling and can be further customized as needed.

```
377  subjectProfileTheme <- function(){  
378    customTheme <- theme_bw() +  
379      theme(panel.grid.major.y = element_line(  
380        colour = c("grey80", theme_bw()$panel.grid$colour))  
381      )  
382    return(customTheme)  
383  }
```

Default Theme in patientProfilesVis

subjectProfileTheme() defines the default theme used across patientProfilesVis outputs.

It is built on top of theme_bw(), one of several predefined ggplot2 themes.

Additional styling is applied through theme() to support patient profile displays.

```
273 gg <- gg + xlim(c(0, 1)) +  
274   subjectProfileTheme() +  
275   theme(  
276     # panel.grid.major = element_blank doesn't  
277     panel.grid.major.x = element_blank(),  
278     panel.grid.minor.x = element_blank(),  
279     panel.grid.major.y = element_blank(),  
280     panel.grid.minor.y = element_blank(),  
281     axis.ticks = element_blank(),  
282     axis.text.x = element_blank(),  
283     axis.ticks.x = element_blank()  
284   )
```

Theme Customization in TEXT Output

subjectProfileTextPlot() extends the default theme for text-based displays.

Grid lines, axis ticks, and axis labels are removed because they add little value to text output.

These adjustments are implemented through an additional theme() layer.

TEXT

What can we customize from the generated output?



Starting Point

Generated directly from the official ADaM example in the patientProfilesVis [vignette](#)

Subject information

Sex, Age

M - 57

Race

WHITE

Planned treatment for period
1

Xanomeline High Dose

Customization Ideas

- Can the text profile be displayed as a table?
- Can the table layout be adjusted?
- Can the variable separator be changed?



Exploration Approach

1. Look for existing functionality
2. Inspect how the output is constructed
3. Make the smallest change that solves the problem
4. Rebuild selected components if needed



LIVE DEMO

Let's walk through the example
pp_demo_adam_text.R

TEXT

What changed after customization?

 **YOUR TURN**
~5 min

Original Output



Subject information

Sex, Age

M - 57

Race

WHITE

Planned treatment for period
1

Xanomeline High Dose



Customized Output



Subject information

Sex

Race

Planned treatment for period 1

M

WHITE

Xanomeline High Dose

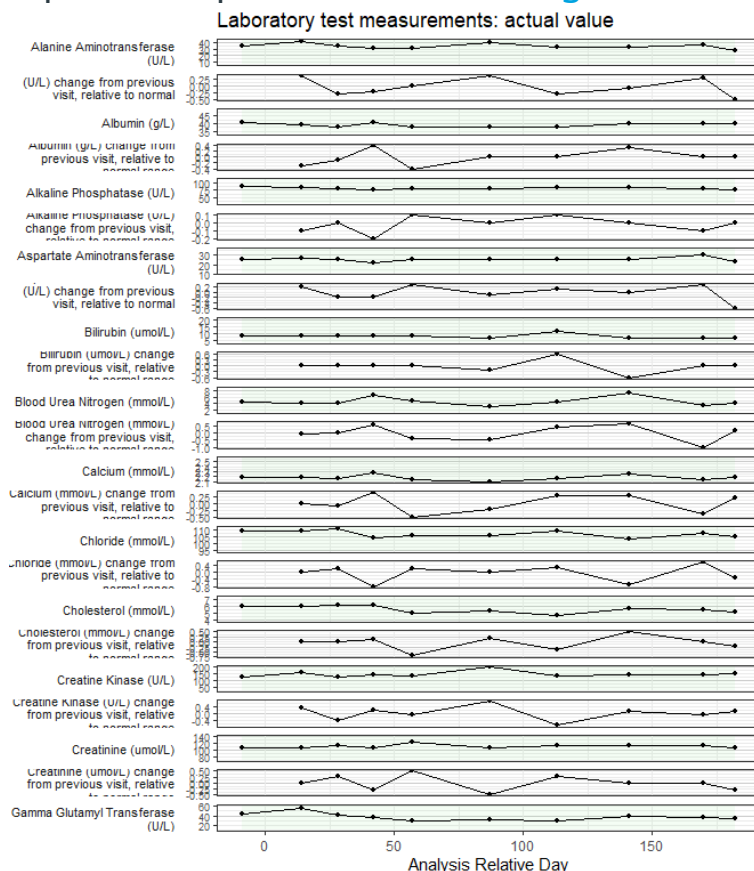
LINE

How can we make the starting point easier to explore?



Starting Point

Generated directly from the official ADaM example in the patientProfilesVis [vignette](#)



Customization Ideas

- Can we select fewer parameters for a clearer view?
- How does paramGroupVar organize parameters in the plot?
- How are long y-axis labels wrapped automatically?



Exploration Approach

1. Look for existing functionality
2. Inspect how the output is constructed
3. Make the smallest change that solves the problem
4. Rebuild selected components if needed



LIVE DEMO

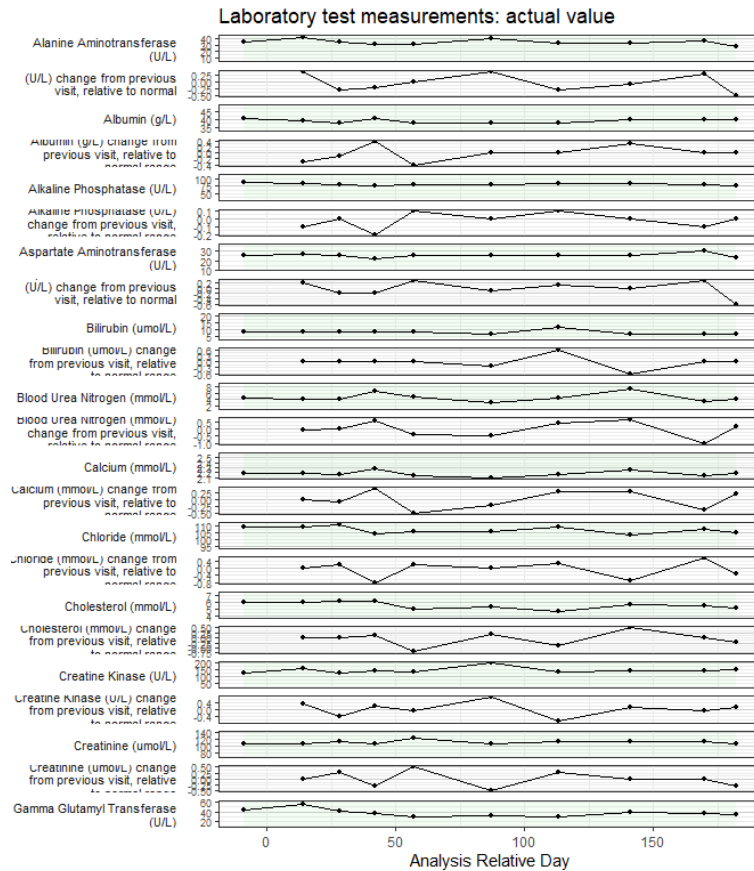
Let's walk through the example
pp_demo_adam_line.R

LINE

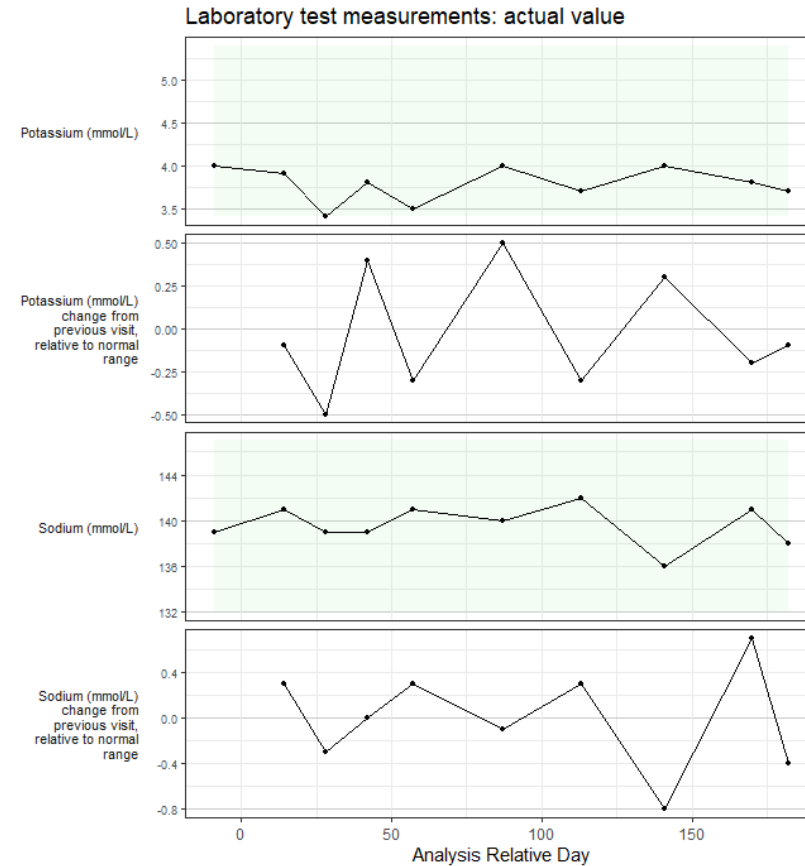
What changed after customization?



Original Output



Customized Output



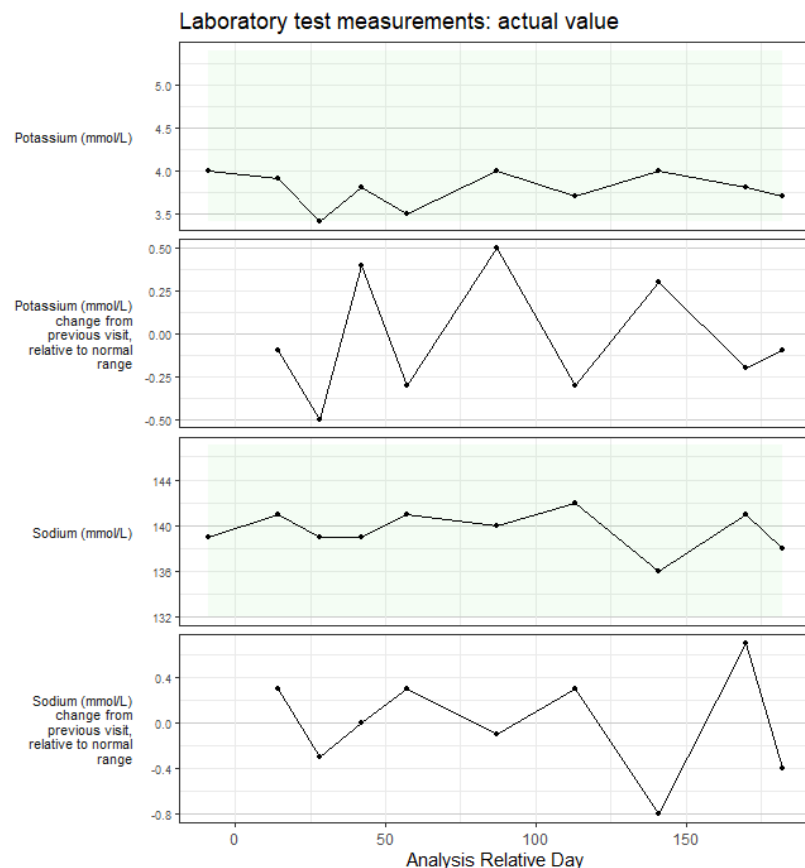
LINE

How can we customize common graphical elements?



Starting Point

Continue from the customized output in the previous exercise



Customization Ideas

- How can we customize point colors?
- How can we customize point shapes?
- How can we add reference lines?
- Can x-axis tick labels be displayed in every panel?



Exploration Approach

1. Look for existing functionality
2. Inspect how the output is constructed
3. Make the smallest change that solves the problem
4. Rebuild selected components if needed



LIVE DEMO

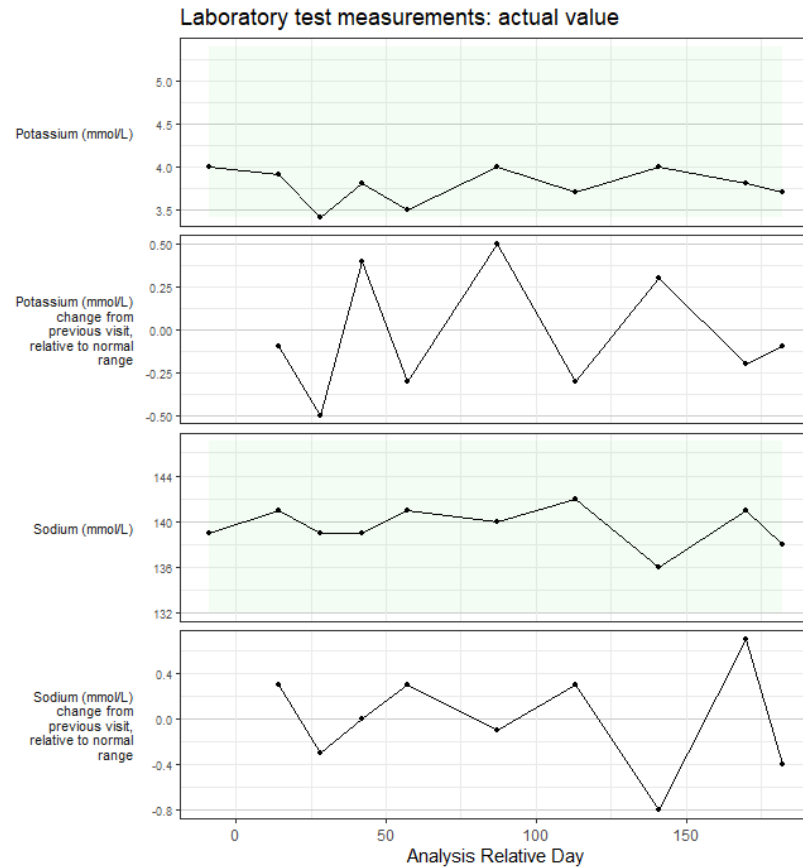
Let's walk through the example
`pp_demo_adam_line.R`

LINE

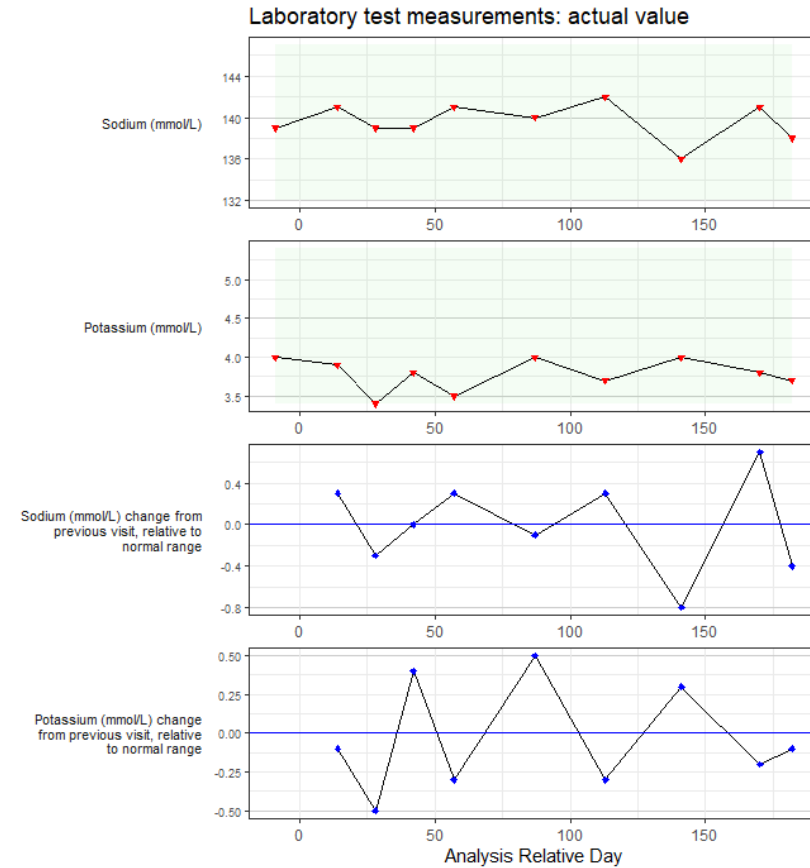
What changed after customization?



Original Output



Customized Output



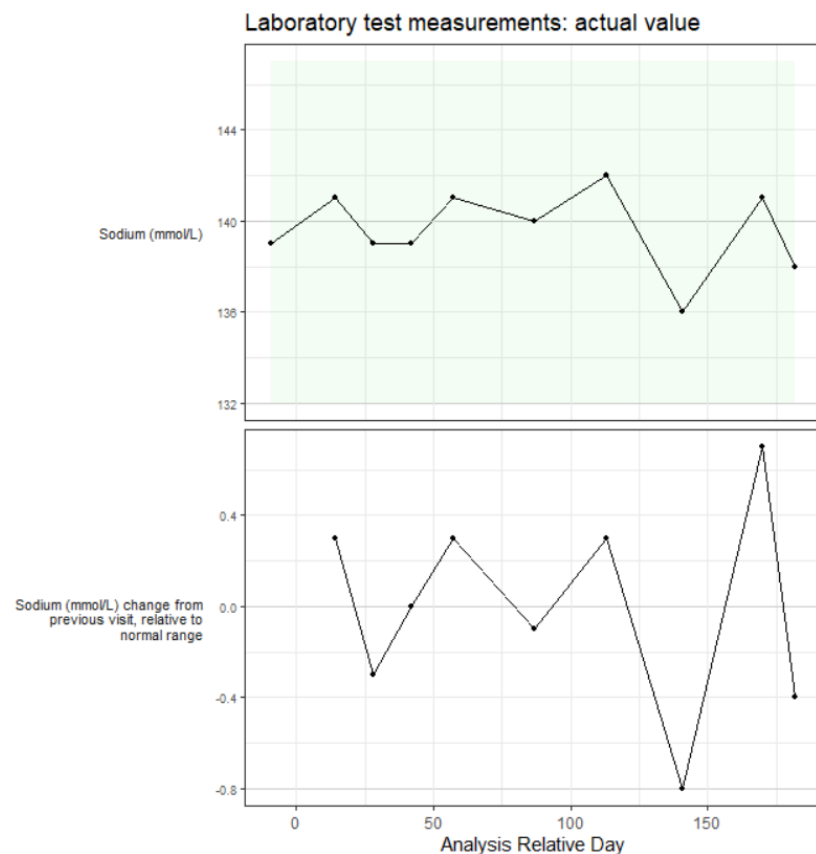
LINE

What if the existing output cannot be adapted easily?



Starting Point

Focus on two parameters from the original [vignette](#) example



Customization Ideas

Can each parameter have axis ranges and tick values suited to its own data range?



Exploration Approach

1. Look for existing functionality
2. Inspect how the output is constructed
3. Make the smallest change that solves the problem
4. Rebuild selected components if needed



LIVE DEMO

Let's walk through the example
`pp_demo_adam_line.R`

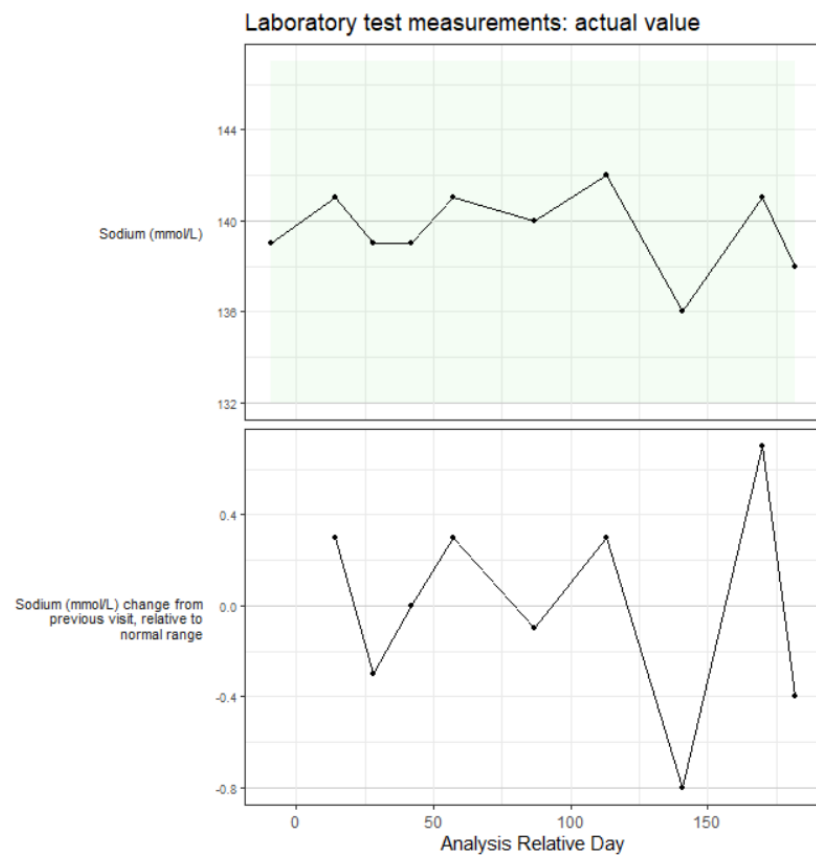
LINE

What changed after customization?

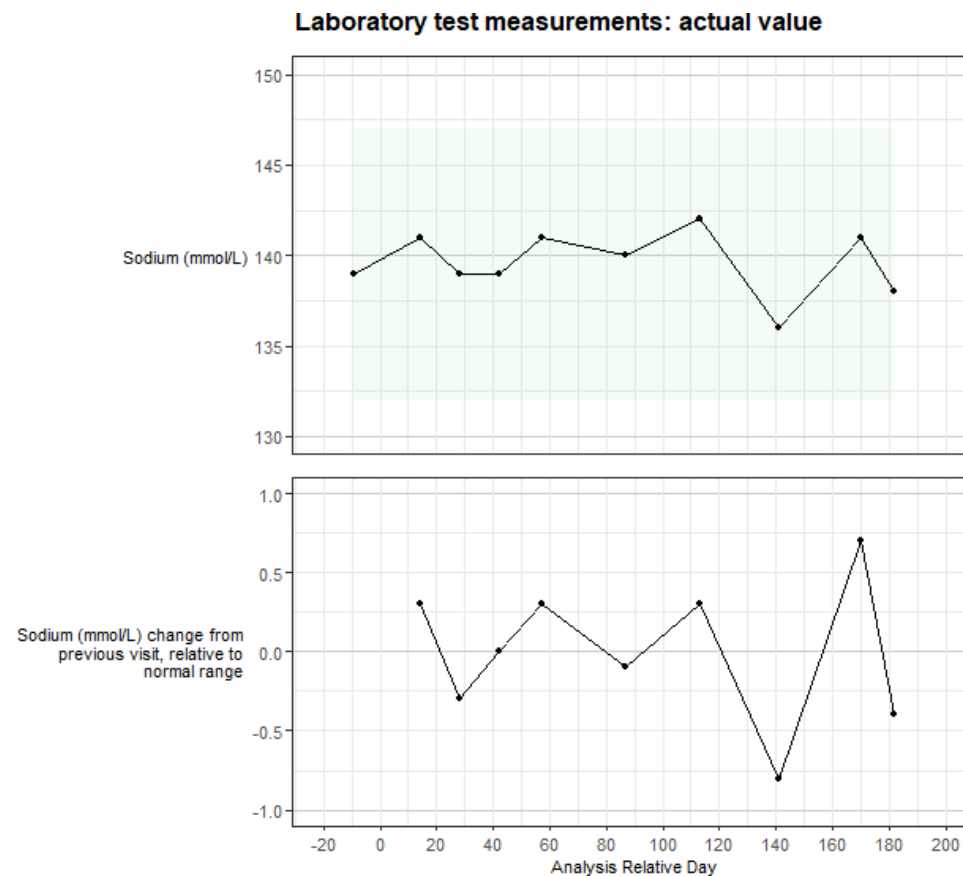
 **YOUR TURN**
~10 min



Original Output



Customized Output



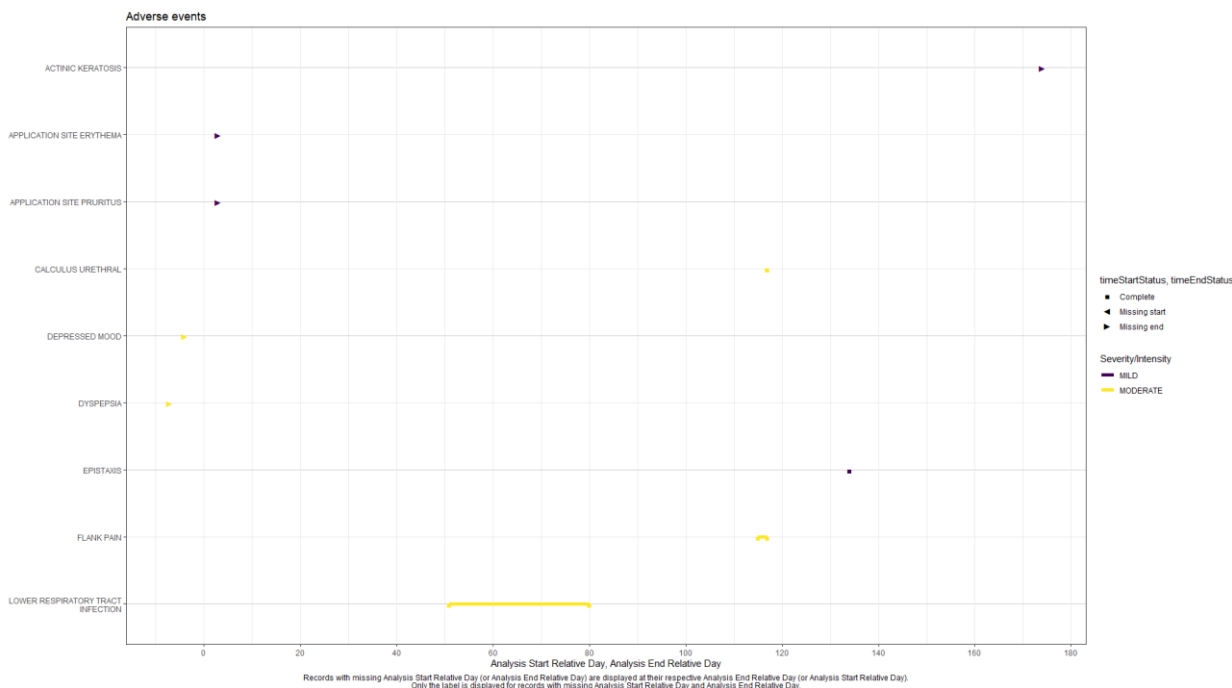
INTERVAL

How can we handle interval data and refine the display?

Starting Point



Adapted from the patientProfilesVis ADaM [vignette](#) example for one selected subject



Customization Ideas

- How are missing interval start/end dates imputed?
- How are multiple periods with different severity levels displayed for the same adverse event?
- How can we control the legend position and displayed levels?
- How can we customize the x-axis tick labels?



Exploration Approach

1. Look for existing functionality
2. Inspect how the output is constructed
3. Make the smallest change that solves the problem
4. Rebuild selected components if needed



LIVE DEMO

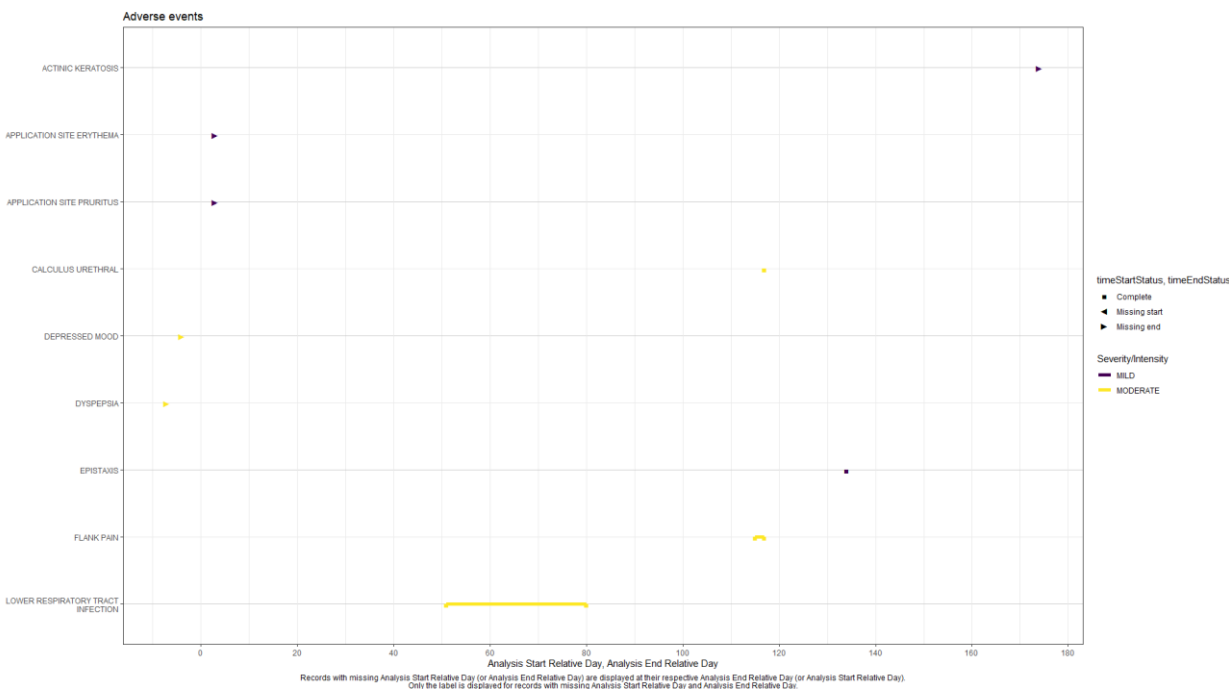
Let's walk through the example `pp_demo_adam_interval.R`

INTERVAL

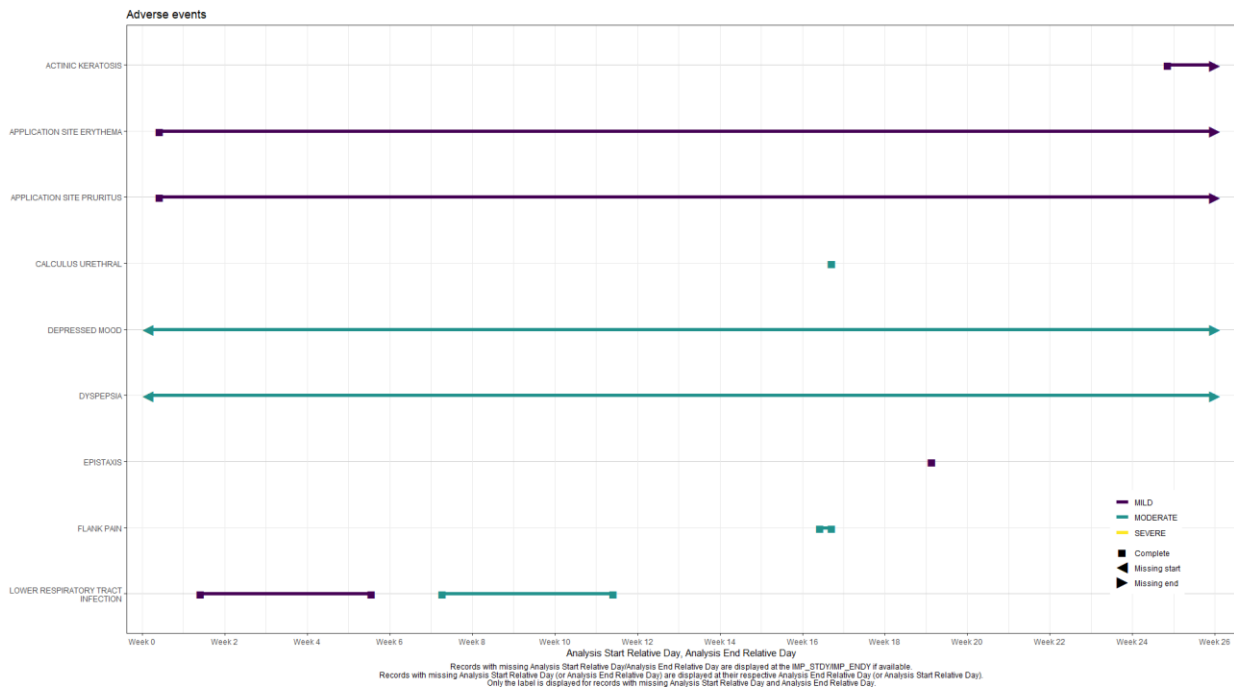
What changed after customization?

 YOUR TURN
~10 min

Original Output



Customized Output



SHINY

 YOUR TURN
~10 min

How can we extend static patient profiles into an interactive application?



Basic Interactive Profile

Adapted from the patientProfilesVis ADaM [vignette](#) example

- Retain the outputs for all subjects across profile modules
- Select a subject interactively from the available profiles
- View text, line, and interval outputs together on a single page



LIVE DEMO

Let's walk through the example
pp_demo_adam_shiny.R



Enhanced Interactive Profile

Extend the basic application with additional content and controls

- Add subject-level text content
- Sort and select parameters
- Reposition the interval legend
- Consider alignment across time-based modules



LIVE DEMO

Let's walk through the example
pp_demo_adam_shiny_enhanced.R

Journey Beyond the Workshop

From Static Graphics to Interactive Review

- Key Takeaways
 - Start with a visualization goal
 - Learn from existing solutions
 - Inspect how the outputs are constructed
 - Choose the simplest approach that meets the need
- Continue Exploring
 - [Workshop Materials](#)
 - [patientProfilesVis CRAN package and vignette](#)
 - [Introduction to ggplot2](#)
 - [Mastering Shiny](#)



Learn new tools by solving real problems

Start with what works, understand how it is built, and extend it when needed